GCSE CHEMISTRY

Higher Tier – Paper 2 style

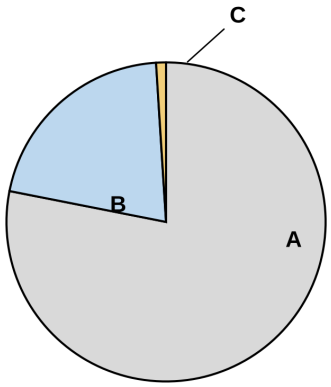
Topic 4.9: Chemistry of the atmosphere

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 115 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **108**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 108

0 1 The pie chart shows the composition of the Earth's atmosphere today. The three sections are labelled **A**, **B** and **C**.



0 1 . 1 Name gas **A** and gas **B**. **[2 marks]**

0 1 . 2 Section C contains several gases. Name **two** of these gases. **[2 marks]**

0 1 . 3 What is the approximate percentage of gas **A** in the air? **[1 mark]**

0 1 . 4 A student uses 100 cm^3 of air and passes it over heated copper. The copper reacts with the oxygen. All the oxygen is used up. **[2 marks]**
What volume of gas should be left when the apparatus has cooled?

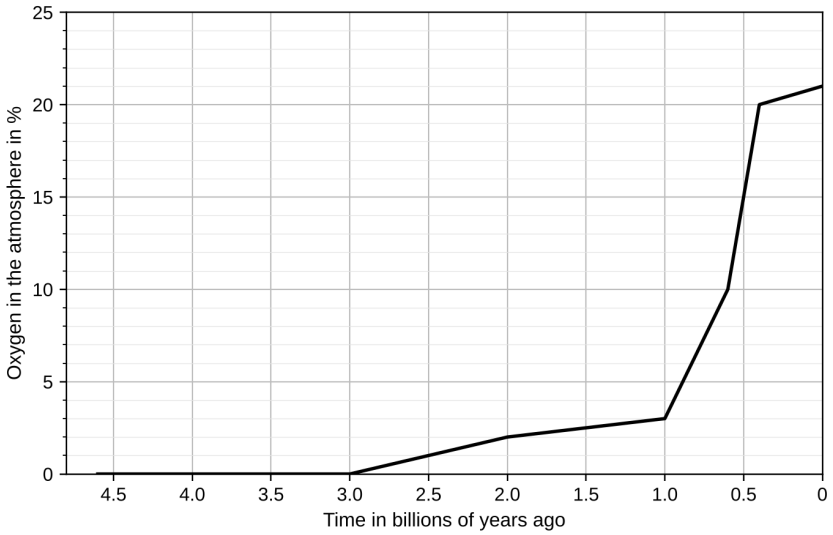
Answer = _____ cm^3

0 1 . 5 The student's apparatus contains 100 cm^3 of air. After the experiment 78 cm^3 remains. Calculate the percentage of oxygen the student measured. **[2 marks]**

Answer = _____ %

0 1 . 6 Write a word equation for the reaction between copper and oxygen. **[1 mark]**

0 2 The graph shows how the percentage of oxygen in the Earth's atmosphere has changed. The Earth is about 4.6 billion years old.



0 2 . 1 Use the graph to estimate when the percentage of oxygen first began to increase. **[1 mark]**

0 2 . 2 What was the percentage of oxygen 0.6 billion years ago? **[1 mark]**

0 2 . 3 Explain how oxygen came to be in the atmosphere. **[2 marks]**

0 2 . 4 Write the symbol equation for photosynthesis. **[2 marks]**

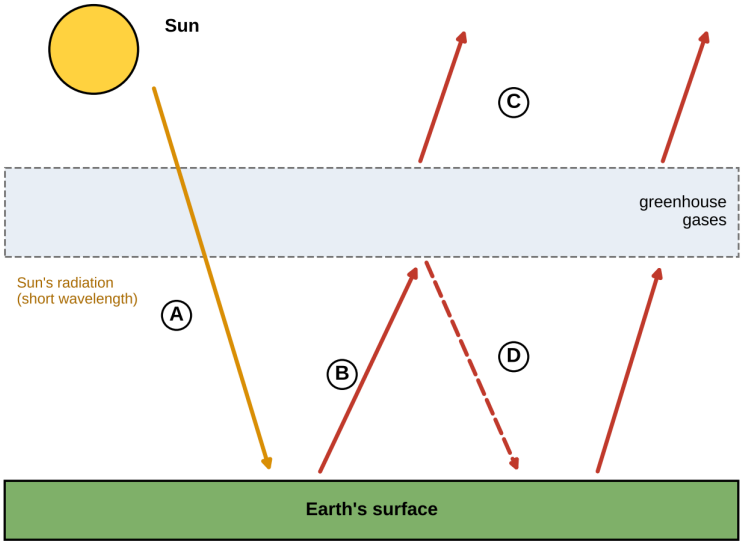
0 2 . 5 Describe the Earth's early atmosphere about 4 billion years ago and how it formed. **[3 marks]**

0 2 . 6 Explain how the oceans formed. **[2 marks]**

0 2 . 7 The amount of carbon dioxide in the atmosphere decreased over millions of years. Give **three** reasons why. **[3 marks]**

0 2 . 8 Scientists are not certain about the Earth's early atmosphere. Suggest why. **[1 mark]**

0 3 The diagram shows how some greenhouse gases keep the Earth warm. The letters **A** to **D** show radiation.



0 3 . 1 Name **two** greenhouse gases. **[2 marks]**

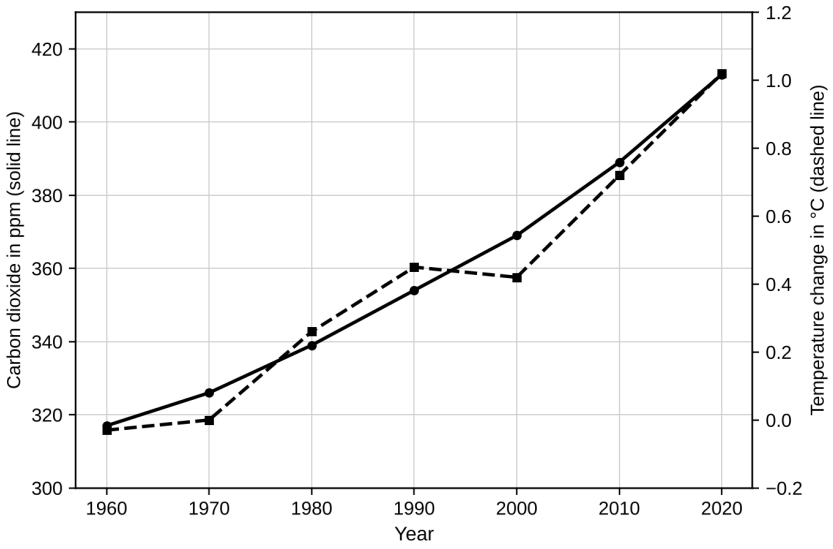
0 3 . 2 Describe how greenhouse gases keep the Earth's surface warm. Refer to the letters **A** to **D** in your answer. **[4 marks]**

0 3 . 3 Give **two** human activities that increase the amount of carbon dioxide in the atmosphere. **[2 marks]**

0 3 . 4 Give **two** human activities that increase the amount of methane in the atmosphere. **[2 marks]**

0 3 . 5 Explain why deforestation increases the amount of carbon dioxide in the atmosphere. **[2 marks]**

0 4 The graph shows the concentration of carbon dioxide in the atmosphere and the change in average global temperature between 1960 and 2020.



0 4 . 1 What was the concentration of carbon dioxide in 1960 and in 2020? **[2 marks]**

0 4 . 2 Calculate the percentage increase in carbon dioxide concentration from 1960 to 2020. **[3 marks]**

Give your answer to 2 significant figures.

Answer = _____ %

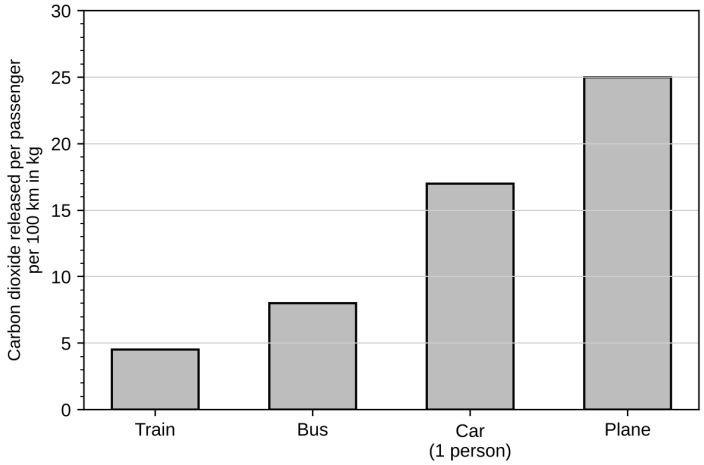
0 4 . 3 Describe the relationship between carbon dioxide concentration and temperature shown by the graph. **[2 marks]**

0 4 . 4 A student says: 'The graph proves that carbon dioxide causes global warming.' Evaluate the student's statement. **[2 marks]**

0 4 . 5 Give **three** possible effects of global climate change. **[3 marks]**

0 4 . 6 It is difficult to predict the effects of climate change accurately. Suggest **one** reason why. **[1 mark]**

0 5 The bar chart shows the carbon dioxide released per passenger for 100 km of travel using four types of transport.



0 5 . 1 Which type of transport produces the most carbon dioxide per passenger? **[1 mark]**

0 5 . 2 How many times more carbon dioxide is released per passenger by the plane than by the train? **[2 marks]**
Give your answer to 2 significant figures.

Answer = _____

0 5 . 3 What is meant by a **carbon footprint**? **[2 marks]**

0 5 . 4 Suggest **two** ways to reduce the carbon footprint of a household. **[2 marks]**

0 5 . 5 A car with 4 people produces 17 kg of carbon dioxide per 100 km. Calculate the carbon dioxide released per passenger. **[1 mark]**

Answer = _____ kg

0 5 . 6 Suggest **two** reasons why it is difficult to reduce carbon dioxide emissions. **[2 marks]**

0 6 This question is about the pollutants made when fuels burn.

0 6 . 1 Complete the table. **[8 marks]**

Pollutant	How it forms	Problem it causes
carbon monoxide		
sulfur dioxide		
oxides of nitrogen		
particulates		

0 6 . 2 Butane (C_4H_{10}) burns completely in oxygen. Write a balanced symbol equation. **[2 marks]**

0 6 . 3 Methane burns in a limited supply of oxygen to make carbon monoxide and water. Write a balanced symbol equation. **[2 marks]**

0 6 . 4 Carbon monoxide is difficult to detect without a special detector. Explain why. **[1 mark]**

0 6 . 5 Explain how acid rain is caused by burning fuels that contain sulfur. **[3 marks]**

0 6 . 6 Give **two** effects of acid rain. **[2 marks]**

0 6 . 7 Suggest **one** way of reducing sulfur dioxide emissions from power stations. **[1 mark]**

0 7 Short answer questions on the atmosphere.

0 7 . 1 Name the gas that makes up most of the Earth's atmosphere. **[1 mark]**

0 7 . 2 Give **one** reason why the amount of oxygen in the atmosphere is currently stable at about 21%. **[1 mark]**

0 7 . 3 Describe how the amount of nitrogen in the atmosphere increased. **[2 marks]**

0 7 . 4 Name **two** noble gases in the air. **[2 marks]**

0 7 . 5 The proportion of carbon dioxide in the early atmosphere was much higher than today. State what happened to the carbon that was in the carbon dioxide. **[2 marks]**

0 8 The table shows the smallest area of sea ice in the Arctic each year at five different dates.

0 8 . 1 Calculate the decrease in the area of sea ice between 1980 and 2020. **[1 mark]**

Year	**1980**	**1990**	**2000**	**2010**	**2020**
Area of sea ice in million km^2	7.7	6.2	6.3	4.6	3.9

Answer = _____ million km²

0 8 . 2 Calculate the percentage decrease in the area of sea ice between 1980 and 2020. Give your answer to 2 significant figures. **[2 marks]**

Answer = _____ %

0 8 . 3 Describe the trend in the data. Mention any anomaly. **[2 marks]**

0 8 . 4 Suggest how the data is linked to the increase in greenhouse gases. **[2 marks]**

0 8 . 5 Explain why melting sea ice does not itself raise sea level much, but melting ice on land does. **[2 marks]**

0 9 Stretch questions (grade 7–9).

0 9 . 1 *Describe how the Earth's atmosphere has changed over the last 4.6 billion years and explain what caused each change. **[6 marks]**

0 9 . 2 *Explain how human activities have affected the Earth's atmosphere, and evaluate the ways of reducing the impact on climate. **[6 marks]**

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.9: Chemistry of the atmosphere

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	A: nitrogen ; B: oxygen ;		2	AO1 4.9.1.1
01.2	argon ; carbon dioxide / water vapour / neon / helium ;		2	AO1 4.9.1.1
01.3	78 – 80 %		1	AO1 4.9.1.1
01.4	oxygen = 21% of 100 = 21 cm^3 ; 100 − 21 = 79 cm^3 ;	Allow 80 cm^3 if 20% is used.	2	AO2 4.9.1.1
01.5	100 − 78 = 22 cm^3 ; 22 % ;		2	AO2 4.9.1.1
01.6	copper + oxygen → copper oxide		1	AO1 4.9.1.1
02.1	about 3 billion years ago (2.5 – 3.0)		1	AO3 4.9.1.2
02.2	10 %		1	AO3 4.9.1.2
02.3	algae and (later) plants evolved ; photosynthesis released oxygen ;		2	AO1 4.9.1.2
02.4	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$; correct formulae and balanced ;	Allow light / chlorophyll above the arrow.	2	AO1 4.9.1.2
02.5	mostly carbon dioxide (with little or no oxygen) ; water vapour, nitrogen and small amounts of methane and ammonia ; from volcanic activity / gases released from the Earth's interior ;	Similar to the atmospheres of Mars and Venus today.	3	AO1 4.9.1.2
02.6	the Earth cooled so water vapour condensed ; forming the oceans ;		2	AO1 4.9.1.2
02.7	dissolved in the oceans ; used by plants and algae in photosynthesis ; locked up in sedimentary rocks (limestone) / fossil fuels (coal, oil, gas) ;	Carbon dioxide dissolving made carbonates that formed sediments.	3	AO2 4.9.1.2
02.8	it happened billions of years ago so there is little evidence / the evidence is limited		1	AO3 4.9.1.2
03.1	carbon dioxide ; methane / water vapour ;		2	AO1 4.9.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	short wavelength radiation from the Sun (A) passes through the atmosphere and is absorbed by the Earth ; the Earth emits longer wavelength infrared radiation (B) ; greenhouse gases absorb some of the infrared radiation ; and re-radiate it in all directions, some back to the Earth (D), warming the surface (the rest escapes to space, C) ;		4	AO1 4.9.2.1
03.3	burning fossil fuels (cars, power stations) ; deforestation / making cement / burning wood ;	Any two.	2	AO1 4.9.2.2
03.4	farming cattle / sheep (digestion) ; rice paddy fields / landfill sites (decay of waste) ;	Any two.	2	AO1 4.9.2.2
03.5	fewer trees to take in carbon dioxide by photosynthesis ; burning / decay of trees releases carbon dioxide ;		2	AO2 4.9.2.2
04.1	317 ppm ; 413 ppm ;	Allow ± 3 ppm.	2	AO3 4.9.2.3
04.2	413 − 317 = 96 ; (96 ÷ 317) × 100 ; 30 % ;	Correct answer with no working scores 3. Allow ecf.	3	AO2 4.9.2.3
04.3	as the carbon dioxide concentration increases the temperature increases ; positive correlation (both rise between 1960 and 2020) ;		2	AO3 4.9.2.3
04.4	the graph shows a correlation, not necessarily a cause ; other factors may affect temperature / more evidence (e.g. models, ice cores, the greenhouse effect) is needed ;		2	AO3 4.9.2.3
04.5	rising sea levels / flooding of coasts ; more extreme weather (storms, droughts) ; changes in rainfall / loss of habitats / changes to food production / species extinction ;	Any three.	3	AO1 4.9.2.3
04.6	computer models are simplified / climate is a very complex system ; some data is unreliable / many variables ;	Any one.	1	AO3 4.9.2.3
05.1	plane		1	AO3 4.9.3
05.2	25 ÷ 4.5 ; 5.6 ;		2	AO2 4.9.3
05.3	the total amount of carbon dioxide (and other greenhouse gases) emitted ; over the full life cycle of a product, service or event ;		2	AO1 4.9.3
05.4	use renewable energy / insulate the house / use less energy ; use public transport / walk / cycle / buy local food ;	Any two sensible ways.	2	AO2 4.9.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	17 ÷ 4 = 4.25 kg (per passenger)		1	AO2 4.9.3
05.6	people may not want to change their lifestyle ; alternatives (renewables, public transport) are expensive / not available everywhere ;	Allow 'lack of international agreement'.	2	AO3 4.9.3
06.1	carbon monoxide: incomplete combustion of fuel (not enough oxygen) ; toxic gas / reduces oxygen carried by haemoglobin (blood) ; sulfur dioxide: burning sulfur impurities in the fuel ; causes acid rain / breathing problems ; oxides of nitrogen: nitrogen and oxygen in the air react at high temperatures in engines ; cause acid rain / respiratory problems ; particulates (soot): incomplete combustion ; global dimming / respiratory problems / damage to lungs ;	One mark for each correct box. Allow 'carbon' for particulates.	8	AO1 4.9.4
06.2	$2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$; correct formulae and balanced ;		2	AO2 4.9.4
06.3	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$; correct formulae and balanced ;	Allow $CH_4 + 1½O_2 \rightarrow CO + 2H_2O$.	2	AO2 4.9.4
06.4	it is colourless and odourless		1	AO1 4.9.4
06.5	sulfur burns to form sulfur dioxide ; sulfur dioxide dissolves in rain water / reacts with water and oxygen ; forming sulfuric acid (acid rain) ;		3	AO2 4.9.4
06.6	damages buildings and statues (limestone) ; kills trees / fish in lakes / harms plants ;	Any two.	2	AO1 4.9.4
06.7	remove sulfur from fuels before burning ; remove sulfur dioxide from waste gases (flue gas desulfurisation) ;	Any one.	1	AO3 4.9.4
07.1	nitrogen		1	AO1 4.9.1.1
07.2	plants / algae produce oxygen by photosynthesis and organisms use it in respiration / combustion (balance)		1	AO3 4.9.1.2
07.3	ammonia was broken down / reacted with oxygen to release nitrogen ; (denitrifying) bacteria also released nitrogen / volcanoes released nitrogen ;		2	AO1 4.9.1.2
07.4	argon ; neon / helium / krypton / xenon ;		2	AO1 4.9.1.1
07.5	it was locked in sedimentary rocks (e.g. limestone) ; and in fossil fuels formed from dead plankton and plants ;		2	AO1 4.9.1.2
08.1	7.7 − 3.9 = 3.8 million km^2		1	AO2 4.9.2.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	3.8 ÷ 7.7 × 100 ; 49 % ;		2	AO2 4.9.2.3
08.3	the area of sea ice decreases over time ; 2000 (6.3) is slightly higher than 1990 (6.2) ;		2	AO3 4.9.2.3
08.4	more greenhouse gases cause the average global temperature to rise ; so more ice melts ;		2	AO3 4.9.2.3
08.5	floating sea ice already displaces its own mass of water ; ice on land adds extra water to the oceans ;		2	AO3 4.9.2.3
09.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • early atmosphere formed by volcanic outgassing: mostly carbon dioxide, with water vapour, nitrogen and traces of methane and ammonia • the Earth cooled and water vapour condensed to form oceans • carbon dioxide dissolved in the oceans and formed carbonates / sediments, reducing its percentage • algae and then plants evolved and used carbon dioxide in photosynthesis, releasing oxygen • carbon was locked up in fossil fuels and sedimentary rocks (limestone) • oxygen increased to about 21% and nitrogen increased to about 78% (released by bacteria, and unreactive so stays); ozone layer formed from oxygen allowing more life on land		6	AO1 4.9.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • burning fossil fuels and deforestation increase carbon dioxide; farming and landfill increase methane • greenhouse gases absorb infrared radiation and re-radiate it back to Earth, increasing average global temperature (enhanced greenhouse effect) • effects include rising sea levels, extreme weather, changes in rainfall and loss of habitats • reducing carbon footprint: renewable energy, energy efficiency, public transport, carbon capture and storage, planting trees • limitations: cost, technology not yet widely available, people reluctant to change lifestyles, need for international cooperation • reasoned judgement e.g. a combination of measures is needed because no single solution is enough		6	AO3 4.9.3

Where marks are lost in Topic 4.9

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Early atmosphere:** volcanoes released carbon dioxide, water vapour, nitrogen, and traces of methane and ammonia. Oxygen came later from photosynthesis. Know the three reasons carbon dioxide fell: oceans, plants, rocks and fossil fuels.
- ☐ **Composition today:** about 80% nitrogen, 20% oxygen (78% and 21% exactly), small amounts of argon, carbon dioxide and water vapour. Use these numbers in calculations.
- ☐ **Greenhouse effect:** radiation from the Sun passes through; the Earth radiates infrared; greenhouse gases *absorb and re-radiate* it back. Do not say they 'trap heat' without saying how.
- ☐ **Correlation is not cause:** a graph showing two things rising together does not prove one causes the other. Mention other evidence.
- ☐ **Carbon footprint:** the total greenhouse gases over the full life cycle of a product, service or event. Reduction ideas should be specific.
- ☐ **Pollutants:** CO and soot from incomplete combustion; SO_2 from sulfur in the fuel; NO_x from nitrogen and oxygen in air reacting at high temperature (not from the fuel).
- ☐ **Percentage change:** (new − old) ÷ old × 100. Give the answer to the number of significant figures asked for.
- ☐ **Acid rain:** sulfur dioxide + water + oxygen → sulfuric acid. Effects: damages limestone buildings, kills trees and aquatic life.